MILITARY SPECIFICATION

CHARGE, PROPELLING, (GREEN BAG), 175MM, M124

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Propelling (Green Bag), M124 for 175MM Gun M113.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	- Powder, Black
MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Proceudres and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

9223106 - Charge, Propelling, 175MM, XM124 for Gun M113

FSC: 1320

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7548187 - Packing and Marking of Metal Containers For Propelling Charges
8857288 - Packing for Container, Metal, Ammunition, XM460, for Charge Propelling, 175MM, M86 for Gun M113
8857378 - Packing, Interim, for Charge, Propelling 175MM, M86, for Gun M113

PUBLICATIONS

ARMY

IEL - 9201183 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Propelling charge assembly.-The propelling charge assembly shall comply with all requirements specified on Drawing (dwg.) 9223106 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder.-The moisture content of black powder at the loading station, at the time of loading, shall not exceed the amount specified in specification MIL-P-223.

3.4 Propelling charge weight.-The propelling charge weight shall comply with the weight specified in the loading authorization within a tolerance of plus or minus 0.40 ounces.

3.5 Profile and alignment.-The propelling charge assemblies shall freely enter a profile and alignment gage (see 4.2.2.6).

3.6 Test plug assembly torque.-The test plug shall be assembled to withstand subsequent application of the minimum assembly torque specified on the applicable drawing, without moving (see 4.2.2.8).

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3.7 Muzzle Velocity.-The corrected muzzle velocity of the projectile at 70° Fahrenheit (°F.) temperature shall be not more tnan 1700 feet per second (fps) nor less than 1650 fps and the standard deviation of the lot shall be not greater than 6.5 fps.

3.8 Workmanship.-All components shall be free of dirt, grease, and other foreign material.

3.9. Initial production test.-The results of this test are for information only.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind

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his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Black powder from not more than one lot from one manufacturer.

b. Propellant from not more than one lot.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be

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used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.2.4.

4.2.2.1 End (see dwg. 9223100 covering a detail of dwg. 9223106)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.0 percent		
201.	Cloth improper color	Visual	01002
202.	Marking missing, misleading or unidentifiable	Visual	01003

4.2.2.2 End, igniter (see dwg. 9223102 covering a detail of dwg. 9223106)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.0 percent		
201.	Cloth improper color	Visual	02002
202.	Marking missing, misleading or unidentifiable	Visual	02003

4.2.2.3 Igniter assembly (see dwg. 9223103 covering a detail of dwg. 9223106)

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Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Any seam incompletley stitched	Visual	03001
102.	Stitching improper	Visual	03002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	03003

4.2.2.4 Igniter loading assembly (see dwg. 9223104 covering a detail of dwg 9223106)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Assembly damaged to the extent that black powder can escape without force	Visual	04001
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Any seam or opening incompletely stitched	Visual	04002
202.	Threads broken	Visual	04003
203.	Evidence of poor workmanship (see 3.8)	Visual	04004

4.2.2.5 Body loading assembly (see dwg. 9223105 covering a detail of dwg. 9223106)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Diameter	Gage	05001
Minor:	AQL 2.5 percent		
201.	Cloth improper color	Visual	05002
202.	Any seam incompletely stitched	Visual	05003

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203.	Thread broken	Visual	05004
204.	Marking missing, misleading or unidentifiable	Visual	05005
205.	Evidence of poor workmanship (see 3.8)	Visual	05006

4.2.2.6 Propelling charge assembly (see dwg. 9223106)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Chamber gage failure (see 3.5)	Gage	06001
102.	Assembly damaged to the extent that propellant can escape without force	Visual	06002
Minor:	AQL 1.0 percent		
201.	Marking missing, misleading or unidentifiable	Visual	06003
202.	Evidence of poor workmanship (see 3.8)	Visual	06004

4.2.2.7 Metal container prior to sealing (see dwg. 8857288 or 8857378)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.5 percent		
201.	Any packing component missing	Visual	07001
202.	Cover cannot be removed by hand	Visual-Manual	07002
203.	Charge cannot be removed by hand	Visual-Manual	07003

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4.2.2.8 Metal container sealed (see dwg. 8857288, 8857378 and 7548187)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Torque test failure of plug (see 3.6)	Test	08001
102.	DOD symbol misleading or unidentifiable	Visual	08002
Minor:	AQL 2.5 percent		
201.	Test plug missing or not greased	Visual	08003
202.	Car seal missing or incorrectly applied	Visual	08004
203.	Cover incompletely engaged	Visual	08005
204.	Marking missing, misleading or unidentifiable	Visual	08006

4.2.3 Testing

4.2.3.1 Moisture content of black powder (see 3.3) - Major defect - Code No. 06005.-The moisture content of the black powder used in loading the igniter loading assembly shall be determined at the time of loading. The contractor shall test at least one sample of black powder from each eight hours production. Any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample. If loading has begun, assemblies loaded with the non-conforming black powder shall be rejected. The test shall be performed as specified in 4.3.1.

4.2.3.2 Propelling charge weight (see 3.4) - Major defect - Code No. 06006, weight of black powder (see dwg. 132688).-Major defect - Code No. 06007.-The charges shall be weighed and check weighed 100 percent. The check weighing shall be accomplished independently of the original weighing using a different balance from that used to make the original weighing and if performed manually, shall be performed by another operator. Any charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

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4.2.3.3 Air pressure test (see dwg. 7548187) - Major defect - Code No. 08008.-The containers shall be checked 100 percent as specified in 4.3.2. If any container fails to meet the applicable requirements, it shall be rejected and removed from the lot.

4.2.3.4 Muzzle velocity (see 3.7) - Major defect - Code No. 06008.-Ten propellant charges shall be randomly selected from each lot for this test. If the sample fails to comply with the velocity requirement or if the standard deviation of the sample multiplied by the factor 0.73 exceeds the applicable lot requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.3.

4.2.3.5 Initial production test.-This test shall be performed on the first production lot. One hundred and fifty six charges shall be selected for this test.

4.2.3.6 Test Validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.2.4 Inspection Equipment.-Index of Inspection Equipment Number IEL-9201183 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.2.4.2 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 Moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with Specification MIL-P-223.

4.3.2 Air pressure test for container.-The container shall be subjected to an internal pressure of 3 pound per square inch using test equipment in accordance with 4.2.4. A pressure drop equivalent to 3 inches of water on a manometer in 15 seconds shall be cause for rejection.

4.3.3 Proving ground test.-This test shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired from a gun for which the cartridge is standard.

4.3.3.1 Muzzle velocity.-The propelling charges shall be assembled into the cartridge and shall be fired at 70 degrees F temperature with inert loaded projectiles and dummy fuzes. The muzzle velocity shall be measured and calculations made for determination of compliance with the applicable requirements.

4.3.4 Initial production test.-The 156 charges shall be subjected to the following test.

4.3.4.1 Drop Test.-Forty-eight of the charges shall be dropped; 24 at minus 65 degrees Fahrenheit and 24 at plus 145 degrees F. Drop each charge five times from a height of seven feet onto armor plate. The drop orientations shall be base down, base down at 45 degrees, base up, base up at 45 degrees, and horizontal.

4.3.4.1.1 Bump Test.-Twenty-four previously dropped charges; 12 at minus 65 degrees F and 12 at plus 145 degrees F. Bump should be conducted on Gaynes Package Tester and should simulate 150 miles over Belgian Block as horizontally stored cargo.

4.3.4.1.2 Bump Test.-Twenty-four previously dropped charges; 12 at minus 65 degrees F and 12 at plus 145 degrees F. Bump should be conducted on Gaynes Tester and should simulate 150 miles over Belgian Block as vertically stored cargo.

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4.3.4.1.3 The 48 previously bumped charges shall be removed from the containers and subjected to unpackaged drop tests; 24 at minus 65 degrees F and 24 at plus 145 degrees F. Drop 15 charges at each temperature one time from a height of five feet onto armor plate. Drop orientations (3 charges per orientation) should be base down, base down at 45 degrees, base up, base up at 45 degrees, and horizontal. Drop 9 charges at each temperature five times from a height of five feet on an armor plate. Drop each charge at each of the five orientations.

4.3.4.2 Vibration test.-Forty-eight of the charges; 24 at minus 65 degrees F and 24 at plus 145 degrees F. Vibration should be in accordance with TECP700-700, Interim Pamphlet 70-73 but the simulation should be conducted with exitation perpendicular to charge axis and then repeated with exitation through the charge axis (charge oriented vertically).

4.3.4.3 Firing.-Sixty charges which have not been subjected to rough handling shall be fired as follows: 15 at minus 65 degrees F, 15 at 0 degrees F including the projectile, 15 at plus 70 degrees F, and 15 at 145 degrees F. Fifteen charges assembled with master standard calibration components and conditioned at plus 70 degrees F shall be fired. The charges fired at zero shall be from a new gun tube.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A.-The propellant charges shall be packed in accordance with dwg. 8857288. Containers shall be closed, sealed, tested and marked in accordance with the requirements of dwg. 7548187.

5.1.2 Level C.-The propellant charges shall be packed in accordance with dwg. 8857378. Containers shall be closed, sealed, tested and marked in accordance with the requirements of dwg. 7548187.

5.2 Marking.-Marking shall be in accordance with dwg. 7548187.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

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b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use

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comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Submission of test data.-In addition to the normal distribution of records, when the charge is procured by the Department of the Army, one copy of all ballistic test data will be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-ND2 and SMUPA-DE2

6.5 Proving Ground test summary

Test	Sample Size	Velocity	Std. Dev.
Velocity	10	see 3.7	see 3.7

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1320-A-165

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